

Part IX — Modern Physics and the Universe · Chapter 51

Relativity Concepts

How space, time, motion, and gravity are bound together into a single fabric

A clock on a GPS satellite runs slightly faster than an identical clock on the ground — not because of a manufacturing defect, but because time itself flows at different rates depending on speed and gravity. Twin astronauts who travel at different speeds age by different amounts, even if they are the same biological age when they depart. Light always travels at exactly the same speed regardless of whether its source is moving toward you or away from you. None of these results make sense in the classical Newtonian framework that served physics for more than two centuries. They all emerge from Albert Einstein's theory of relativity — arguably the most radical and successful reconceptualisation of the physical world in the history of science. This chapter introduces the foundational ideas of both special relativity (1905), which concerns motion at constant velocity, and general relativity (1915), which incorporates gravity and accelerated frames. No tensor calculus is required. The goal is conceptual clarity: to understand why the universe works the way it does at extreme speeds and in strong gravitational fields.

51.1 Why Classical Ideas of Space and Time Failed

For two centuries after Newton, physicists assumed that space and time were fixed, universal, and independent. Space was an absolute arena in which events occurred; time was a river that flowed at the same rate everywhere, for all observers, regardless of their motion. Motion was measured against this fixed background.

Two experiments in the late nineteenth century exposed deep problems with this view. The first was the Michelson–Morley experiment of 1887. At that time, physicists believed that light (like all waves) must travel through a medium — a hypothetical substance called the luminiferous aether that filled all of space. If the aether were stationary, Earth's motion through it should cause light to travel at slightly different speeds in different directions, just as a boat travels faster downstream than upstream. Michelson and Morley built an extraordinarily precise interferometer to detect this speed difference. They found nothing: light speed appeared identical in all directions, regardless of Earth's motion. The aether did not exist.

The second problem was theoretical. James Clerk Maxwell's equations of electromagnetism (Chapter 27) predicted that light travels at a fixed speed $c \approx 3 \times 10^8$ m/s. But fixed with respect to what? Newtonian mechanics said that speeds add: a ball thrown at 10 m/s from a car moving at 20 m/s is moving at 30 m/s relative to the ground. If the same rule applied to light, a flashlight shone from a moving spaceship would emit light at $c + v$ or $c - v$ depending on direction. Maxwell's equations gave no such variation. The two theories were irreconcilable.

Einstein resolved both problems by doing something radical: he took both Maxwell's equations and experiment seriously. If light always travels at c , then c really is constant for all observers. And if c is constant, then space and time cannot both be fixed and absolute. Something had to give. Einstein chose to keep c fixed and revise the concepts of space and time.

Historical Note: Albert Einstein published his special theory of relativity in 1905, at age 26, while working as a patent examiner in Bern, Switzerland. He had no academic position and no laboratory. The paper, "On the Electrodynamics of Moving Bodies," contained no references to other scientists and introduced a conceptual revolution with minimal mathematics. General relativity followed ten years later in 1915, after Einstein spent years grappling with how to incorporate gravity.

51.2 Einstein's Postulates of Special Relativity

Special relativity rests on two postulates — fundamental assumptions from which all the theory's predictions follow by logical reasoning.

The Rule: Postulate 1 — The Principle of Relativity: The laws of physics are the same in all inertial (non-accelerating) reference frames. No experiment performed entirely within a uniformly moving frame can determine that frame's absolute velocity or prove that one frame is "at rest" and another is "moving."

The Rule: Postulate 2 — The Constancy of the Speed of Light: Light travels in a vacuum at the same speed $c \approx 2.998 \times 10^8$ m/s in all inertial reference frames, regardless of the speed of the light source or the observer.

The first postulate is a generalisation of Galileo's principle of relativity (Chapter 3): a ball rolls the same way on a smoothly moving train as it does on a stationary platform. Einstein extended this to all physics, including electromagnetism. The second postulate is the radical one: it contradicts Newtonian velocity addition and forces a revision of how space and time work.

Together, the two postulates mean that any observer — regardless of how fast they are moving — measures light at the same speed c . Two observers moving at different

velocities both measure the same beam of light at c . This seems impossible if space and time are fixed: if both observers measure c for the same beam, and they are moving at different speeds, then something about their measurements of distance and time must differ. Einstein showed that both space (lengths) and time (durations) are observer-dependent quantities.

Key Point: Special relativity applies to inertial frames — reference frames moving at constant velocity. Accelerating frames require general relativity. However, the effects of special relativity (time dilation, length contraction) apply to any inertial observer viewing a moving object, even if that object later accelerates.

51.3 The Constancy of the Speed of Light

The speed of light in vacuum, $c \approx 2.998 \times 10^8$ m/s, is one of the most fundamental constants in nature. It is not merely the speed at which visible light travels: it is the maximum speed at which any information, energy, or causal influence can propagate through the universe. No object with mass can reach or exceed c ; massless particles (photons, gravitons) always travel at exactly c in vacuum.

The universality of c leads to some counterintuitive results. Suppose a spaceship is travelling toward you at $0.9c$ (90% of the speed of light) and its crew fires a laser beam forward. You might expect to measure the laser light at $c + 0.9c = 1.9c$. Instead, you measure exactly c . The crew also measures the laser light at c . This is not an experimental limitation: the relativistic velocity addition formula ensures that no combination of sub-light velocities can ever produce a speed at or above c .

Example — Relativistic velocity addition (conceptual) If Ship A moves at $0.6c$ relative to the ground, and Ship B moves at $0.6c$ relative to Ship A in the same direction, classical physics gives $0.6c + 0.6c = 1.2c$ — faster than light. The relativistic formula gives approximately $0.88c$. The ships add their velocities, but the result is always less than c . Light beams always add to exactly c .

The fact that c is finite and universal has profound implications for astronomy. When we observe a star one hundred light-years away, we see it as it was one hundred years ago: the light left the star one hundred years before arriving at our telescopes. Observing distant galaxies is literally looking back in time. The most distant objects we can observe are so far away that the light reaching us today was emitted less than a billion years after the Big Bang.

51.4 Reference Frames Revisited

A reference frame is a coordinate system plus a set of clocks used to assign positions and times to events. An inertial reference frame is one that is not accelerating — it moves at constant velocity (or is at rest). The laws of physics take the same form in all inertial frames.

The key insight is that there is no preferred inertial frame. No observation made entirely within one inertial frame can reveal whether that frame is “truly at rest” or moving at constant velocity. The surface of Earth is approximately an inertial frame for most purposes. A train moving at constant speed is an equally valid inertial frame. A spaceship drifting far from any gravitating body at constant velocity is an inertial frame.

When we say that a person is “moving at $0.8c$,” we always mean relative to some specified reference frame. From the perspective of the moving person, they are at rest and the rest of the universe is moving past them at $0.8c$. Relativity insists that both perspectives are equally valid. The effects of time dilation and length contraction are always reciprocal: each observer sees the other’s clock running slow and the other’s lengths contracted.

This reciprocity is not a paradox or a contradiction; it is the natural consequence of there being no absolute space or time against which to measure motion. The predictions of special relativity become paradoxical only if one secretly assumes that one frame is special (usually the one the observer is sitting in). Consistently applied, the theory is self-consistent.

51.5 Simultaneity Is Relative

In Newtonian physics, two events that occur at the same time (but at different locations) are simultaneous for all observers. Einstein showed that simultaneity is not absolute: two events that are simultaneous in one reference frame are not simultaneous in a different frame moving relative to the first.

Example — Train thought experiment A train moves at high speed. Lightning strikes both ends of the train simultaneously (in the reference frame of the ground). An observer standing at the midpoint of the platform sees both flashes at the same moment, confirming they were simultaneous. A passenger sitting at the midpoint of the moving train, however, moves toward one flash and away from the other. The passenger receives the front flash before the rear flash. Since the passenger is also at the midpoint of the train, and light travels at c in all frames, the passenger concludes that the front strike happened before the rear strike. Both observers are correct in their own frame. The events are simultaneous in the ground frame but not in the train

frame.

The relativity of simultaneity is not merely a measurement artifact: it means there is no objective, observer-independent “now.” Two spatially separated events that are simultaneous in one frame are genuinely not simultaneous in another frame. The order of some pairs of events can even be reversed depending on the observer’s frame — though only for events that cannot causally influence each other (events separated by more distance than light can travel in the time between them).

This has deep implications. Concepts like “the state of the universe right now” are frame-dependent. The universe does not have a single, globally shared “now.” This is not a limitation of our measuring instruments; it is a fundamental feature of spacetime structure.

Common Mistake: The relativity of simultaneity does not mean “anything goes” or that cause and effect can be violated. Events that are causally connected (one causes the other) are always in the same temporal order in every reference frame. Only events that are spacelike-separated (no causal connection possible) have frame-dependent ordering. Relativistic causality is perfectly preserved.

51.6 Time Dilation

Time dilation is perhaps the most famous and counterintuitive prediction of special relativity: a moving clock runs slower than an identical stationary clock. The faster the clock moves relative to an observer, the slower it ticks as measured by that observer.

The Lorentz factor γ (gamma) quantifies this effect. For an object moving at speed v relative to an observer:

$$\gamma = 1 / \sqrt{1 - v^2/c^2}$$

The Lorentz factor γ . When $v = 0$, $\gamma = 1$ (no dilation). As $v \rightarrow c$, $\gamma \rightarrow \infty$. At $v = 0.87c$, $\gamma \approx 2$ (clocks run at half speed); at $v = 0.995c$, $\gamma \approx 10$.

$$\Delta t = \gamma \Delta t_0$$

Time dilation formula. Δt_0 = proper time (time elapsed on the moving clock as measured in its own rest frame); Δt = dilated time (time elapsed as measured by the stationary observer). Since $\gamma \geq 1$, $\Delta t \geq \Delta t_0$: the stationary observer measures more time.

The proper time is the time measured by a clock that is present at both events (start and end of the interval), i.e., a clock that travels with the moving object. It is always the shortest time interval any observer can assign to that pair of events. All other observers measure a longer time interval.

Example — Muon decay Muons are subatomic particles produced in the upper

atmosphere when cosmic rays strike air molecules, at an altitude of about 15 km. Muons at rest have a half-life of about $1.5 \mu\text{s}$. Travelling at $0.998c$, a muon classically should travel only $0.998c \times 1.5 \times 10^{-6} \text{ s} \approx 450 \text{ m}$ before decaying — far short of 15 km. Yet muons are routinely detected at the Earth's surface. Time dilation explains this: at $v = 0.998c$, $\gamma \approx 16$. The muon's internal clock runs 16 times slower relative to the ground. Its effective ground-frame half-life is $16 \times 1.5 \mu\text{s} = 24 \mu\text{s}$, long enough to reach the surface. Muon detection is direct experimental confirmation of time dilation.

Speed (v/c)	Lorentz Factor (γ)	Moving Clock Rate	Example
0.10	1.005	99.5% of normal	Low-Earth orbit satellites (approx)
0.50	1.155	86.6% of normal	Fast particles in lab experiments
0.87	2.00	50% of normal	Clock runs at half speed
0.95	3.20	31% of normal	High-energy cosmic ray protons
0.99	7.09	14% of normal	Particles in large accelerators
0.999	22.4	4.5% of normal	Near-light-speed travel scenarios
1.000	∞	0% (stopped)	Only massless particles (photons)

Table 51.1 Lorentz factor and time dilation at various speeds. Moving clocks always tick slower as measured by a stationary observer.

Study Tip: When solving time dilation problems: identify which time is the proper time (Δt_0 , measured by a clock travelling with the moving object). The dilated time (Δt) is always larger. If you are unsure which is which, ask: which observer is present at both the start and end of the event? That observer measures proper time.

51.7 Length Contraction

Length contraction is the companion effect to time dilation: an object in motion appears shorter (in the direction of motion) to a stationary observer than it does to an observer moving with the object. The object's length measured in its own rest frame is called its proper length L_0 ; the length measured by a stationary observer is:

$$L = L_0 / \gamma = L_0 \sqrt{1 - v^2/c^2}$$

Length contraction. $L < L_0$: the moving object appears shorter. Only the dimension parallel to the motion is contracted; perpendicular dimensions are unchanged.

A spaceship of proper length 100 m travelling at $0.87c$ ($\gamma = 2$) appears to a stationary observer as only 50 m long. Its width and height are unchanged. The crew of the spaceship do not feel compressed: from their perspective, the ship is its full 100 m length, and it is the external universe that appears contracted in the direction of travel.

Example — Muon decay (length contraction perspective) In the muon's rest frame, the muon is not time-dilated — it lives its normal $1.5 \mu\text{s}$ half-life. But from the muon's perspective, the atmosphere is moving toward it at $0.998c$ and is therefore length-contracted by $\gamma \approx 16$. The 15 km of atmosphere appears only $15\,000/16 \approx 940$ m thick. The muon easily crosses this compressed distance in its normal lifetime. Both the time dilation explanation (ground frame) and the length contraction explanation (muon frame) predict the same result: muons reach the surface.

Length contraction and time dilation are not separate, independent effects: they are two aspects of the same underlying geometry of spacetime. An observer moving relative to you sees your clocks running slow and your lengths contracted; you see their clocks running slow and their lengths contracted. Both views are consistent within their respective frames.

Common Mistake: Length contraction applies only to the dimension along the direction of motion. A spaceship moving left-right is contracted left-right but not up-down or front-back. Also: the object does not physically compress; its atoms are not closer together in its own frame. Only the measurements made by observers in relative motion differ.

51.8 Relativistic Mass-Energy Concepts (Conceptual)

Perhaps the most famous equation in physics, $E = mc^2$, emerges from special relativity. It states that mass and energy are equivalent: mass is a form of energy and energy has the equivalent of a mass. The c^2 factor ($\approx 9 \times 10^{16}$ J/kg) is enormous, which means even a tiny amount of mass corresponds to a vast amount of energy.

$$E = mc^2$$

Rest energy of an object with mass m . This is the energy equivalent of the mass when the object is at rest. For a moving object, the total energy is $E = \gamma mc^2$, which includes kinetic energy. As $v \rightarrow c$, the kinetic energy $\rightarrow \infty$: infinite energy would be required to accelerate a massive object to c .

This equation has several important consequences. First, it explains why nuclear reactions release so much more energy than chemical reactions: nuclear reactions convert a small but non-negligible fraction of the mass of nuclei directly into energy (binding energy changes). A chemical reaction rearranges electrons and converts energy of a few eV per atom; a nuclear reaction can release millions of eV per nucleus by converting mass fractions of approximately 0.1%.

Second, $E = mc^2$ implies that as an object is accelerated toward c , the energy required to accelerate it further increases without limit ($\gamma \rightarrow \infty$). An object with mass cannot be accelerated to c ; it would require infinite energy. Only massless particles, which always travel at exactly c , can reach that speed. This is why c is the cosmic speed limit.

Third, the rest energy mc^2 is always present in a massive object, even at rest. It is not a kinetic energy — it is the intrinsic energy associated with the existence of the mass. This energy can be released (partially) in nuclear fission, fusion, and matter-antimatter annihilation.

Historical Note: The first direct test of $E = mc^2$ came in 1932 when Cockcroft and Walton in Cambridge used a particle accelerator to split lithium nuclei. The mass of the products was measurably less than the mass of the reactants, and the difference corresponded precisely to the kinetic energy released, consistent with $\Delta E = \Delta mc^2$. They shared the 1951 Nobel Prize in Physics for this work.

51.9 Spacetime and the Union of Space and Time

The deepest insight of special relativity is that space and time are not separate entities. They are aspects of a single four-dimensional continuum called spacetime, first described mathematically by Hermann Minkowski in 1908.

In ordinary Euclidean space, the distance between two points is calculated using the Pythagorean theorem and is the same for all observers. In spacetime, the analogous quantity is the spacetime interval, which combines spatial distance and time interval. The spacetime interval between two events is the same for all inertial observers — even though they may disagree about the spatial distance and time between the events separately.

This is the key: different observers “slice” spacetime differently. What one observer calls a purely spatial distance, another observer (moving relative to the first) sees as a mixture of spatial distance and time. What one observer calls a time interval, another sees as involving some spatial component as well. The individual measurements (space, time) are frame-dependent, but the spacetime interval is invariant.

Spacetime diagrams (also called Minkowski diagrams) are a useful visual tool. Time is plotted vertically, one spatial dimension horizontally. An object at rest traces a vertical line (worldline) through spacetime. An object in motion traces a tilted worldline. A light beam traces a 45-degree line (if the axes are scaled appropriately). Events outside the light cone of a given event cannot causally influence it; only events inside the past light cone could have caused it, and only events inside its future light cone can be influenced by it.

“Henceforth space by itself and time by itself are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality.” — Hermann Minkowski, 1908

Key Point: The unification of space and time into spacetime is not a metaphor or an approximation: it is the geometric structure of the universe. When physicists talk about “curved spacetime” in general relativity, they mean that this four-dimensional fabric is genuinely curved by the presence of mass and energy — with measurable physical consequences.

51.10 Gravity in General Relativity (Conceptual Overview)

Special relativity describes physics in inertial (non-accelerating) frames. But real physics includes acceleration and gravity. Einstein spent ten years after 1905 extending his theory to cover these cases. The result, general relativity (GR), published in 1915, is one of the most mathematically sophisticated and physically accurate theories in all of science.

The central insight of general relativity is the equivalence principle: there is no local experiment that can distinguish between being in a gravitational field and being uniformly accelerated. An astronaut in a rocket accelerating at 9.81 m/s^2 far from any star feels exactly the same weight as standing on Earth's surface. A person in free fall near Earth is locally equivalent to floating in empty space (the classic "weightlessness" of orbit).

This principle led Einstein to the radical conclusion that gravity is not a force in the Newtonian sense at all. Gravity is the curvature of spacetime caused by the presence of mass and energy. Objects in free fall are not being pushed or pulled; they are following the straightest possible paths (geodesics) through curved spacetime. The Earth does not pull the Moon; the Earth curves spacetime, and the Moon follows the straightest path through that curved geometry.

Example — Elevator thought experiment Einstein imagined a person in a sealed elevator. If the elevator is at rest on Earth, they feel their normal weight. If the elevator is accelerated upward in empty space at g , they feel the same weight. If the elevator is in free fall on Earth, they feel weightless. If the elevator drifts in empty space far from gravity, they feel weightless. Gravity and acceleration are locally indistinguishable. This equivalence, elevated to a principle, implies that gravity must affect light and time in the same way that acceleration does.

51.11 Curved Spacetime

General relativity describes gravity as curvature of the four-dimensional spacetime fabric. Mass and energy tell spacetime how to curve; the curvature of spacetime tells matter how to move. This two-way interaction is described by Einstein's field equations — a system of ten coupled partial differential equations that are among the most beautiful and complex in physics.

A useful (though imperfect) analogy is a heavy ball placed on a stretched rubber sheet. The ball creates a depression — a curvature — in the sheet. A smaller marble rolling across the sheet follows a curved path, not because a force pulls it toward the heavy ball, but because it follows the contour of the curved surface. In reality, it is not a two-dimensional sheet curving in three dimensions, but a four-dimensional spacetime curving in ways that have no simple visual analogy.

One direct consequence of spacetime curvature is gravitational lensing: the path of light is bent as it passes near a massive object. Einstein predicted that the Sun's gravity should bend light from distant stars passing near the solar limb. In 1919, British astronomer Arthur Eddington led an expedition to observe a total solar eclipse and measured precisely this bending, confirming general relativity and making Einstein world-famous overnight.

Another consequence is gravitational waves — ripples in spacetime itself, propagating outward from accelerating massive objects (such as merging black holes or neutron stars). Einstein predicted gravitational waves in 1916; they were directly detected for the first time in 2015 by the LIGO collaboration, a century later, confirming another major prediction of general relativity.

Historical Note: The 1919 Eddington expedition measured the deflection of starlight by the Sun during a solar eclipse. The measured deflection of approximately 1.75 arcseconds matched Einstein's general relativistic prediction exactly and was twice the deflection predicted by naive Newtonian gravity. The result made front-page news worldwide and established Einstein as the preeminent scientist of his era.

51.12 Gravitational Time Dilation

General relativity predicts not only that relative motion slows clocks (special relativistic time dilation) but also that gravity slows clocks. A clock in a strong gravitational field runs slower than an identical clock in a weaker gravitational field. This is gravitational time dilation.

The deeper the gravitational potential well, the slower time runs. A clock on the surface of Earth runs slightly slower than a clock at high altitude. A clock near the surface of a neutron star (enormously stronger gravity) would run dramatically slower. Clocks at the event horizon of a black hole (the point of no return) effectively stop relative to distant observers.

Example — GPS and gravitational time dilation GPS satellites orbit at about 20 200 km altitude. Two relativistic effects act on their clocks simultaneously. (1) Special relativity: the satellites move at about 3.9 km/s relative to the ground, causing their clocks to run slow by about 7 microseconds per day due to velocity. (2) General relativity: the satellites are in a weaker gravitational field than clocks on the ground, so their clocks run fast by about 45 microseconds per day due to gravitational time dilation. The net effect is +38 microseconds per day. If uncorrected, GPS positions would accumulate errors of about 10 km per day. GPS engineers correct for both effects, providing one of the most precise everyday applications of general relativity.

Gravitational time dilation is not a theory or an approximation in GPS systems: it is a measured, corrected, and essential feature of navigation technology. Every GPS calculation you perform is evidence that general relativity is correct.

51.13 Relativity in Modern Technology

Relativity is not an abstract academic theory: it has direct engineering consequences in several modern technologies.

Global Positioning System (GPS)

As described in Section 51.12, GPS satellites require corrections for both special and general relativistic time dilation. Without these corrections, the system would be useless for navigation within hours.

Particle Accelerators

The Large Hadron Collider at CERN accelerates protons to $0.999999991c$, giving Lorentz factors of over 7000. Engineers use relativistic mechanics, not Newtonian, to design the magnetic fields that steer these particles. Proton lifetime dilation allows these particles to circulate for hours despite having nanosecond rest-frame half-lives. All accelerator physics is inherently relativistic.

Positron Emission Tomography (PET Scans)

PET scanners rely on matter-antimatter annihilation: a positron (antimatter electron) annihilates with an electron, converting their combined rest mass entirely into energy (two gamma ray photons), as predicted by $E = mc^2$. Detecting these photons allows precise imaging of metabolic activity in the brain and body.

Nuclear Power and Weapons

Nuclear fission and fusion convert a fraction of nuclear mass into energy via $E = mc^2$. In uranium-235 fission, approximately 0.09% of the mass is converted to energy — releasing about 200 MeV per fission event, roughly a million times more energy per unit mass than chemical combustion.

Gravitational Wave Detectors

The LIGO and Virgo observatories detect spacetime ripples from merging black holes and neutron stars billions of light-years away. These instruments measure changes in length smaller than one ten-thousandth the diameter of a proton — among the most precise measurements in human history. Gravitational wave astronomy has opened an entirely new observational window on the universe.

51.14 Common Misunderstandings About Relativity

- "Everything is relative" — Relativity does not say that all measurements are equally valid or that truth is subjective. Many quantities are frame-dependent (simultaneity, time intervals, lengths), but others are invariant (spacetime intervals, the speed of light, physical laws). Relativity establishes what is absolute and what is relative, replacing an incorrect absolute with a more accurate one.
- Time dilation means time slows down for the moving person — Incorrect. The moving person experiences proper time normally. Time dilation is always what an observer sees when looking at a clock in relative motion. From the moving clock's perspective, it is keeping perfect time and the outside clock is running slow.
- "The Twin Paradox" means relativity is contradictory — In the Twin Paradox, one twin travels at high speed and returns younger than the stay-at-home twin. This seems contradictory if both twins should see the other as time-dilated. The resolution: the travelling twin accelerates (turns around), breaking the symmetry. The situation is not symmetric; the non-inertial twin ages less. There is no contradiction.
- Relativity only matters near the speed of light — Gravitational time dilation is significant even at low velocities if the gravitational field is strong enough. GPS, atomic clocks, and precision navigation must account for relativistic effects at orbital speeds well below $0.01c$.
- General relativity replaced Newton's gravity and made it wrong — Newtonian gravity is an excellent approximation in weak gravitational fields at low velocities. General relativity reduces to Newtonian gravity in these limits. Scientists use Newtonian gravity for most solar system applications; GR is required only where precision demands it or fields are very strong (near black holes, neutron stars, or for accurate GPS).
- Nothing can travel faster than light — (applied incorrectly) The cosmic speed limit applies to the transmission of information and the motion of matter and energy through space. The universe's expansion can separate two points of space faster than light without violating relativity, because spacetime itself is expanding, not matter moving through space. Quantum entanglement correlations also propagate "faster than light" but cannot transmit information faster than c .

Common Mistake: Relativistic effects such as time dilation and length contraction are frame-dependent: each observer sees the other's clock as slow and the other's lengths as contracted. This is not a paradox — it is the correct prediction. Paradoxes arise only from hidden assumptions about an absolute or preferred frame. Eliminating those

assumptions resolves every apparent contradiction.

Chapter Summary

Relativity — both special (1905) and general (1915) — reveals that space, time, mass, and gravity are interconnected in ways invisible at everyday speeds but essential at extreme speeds and in strong gravitational fields.

- Classical mechanics assumed absolute space and time; the Michelson–Morley experiment and the constancy of light speed showed this was wrong.
- Special relativity rests on two postulates: the equivalence of inertial frames, and the constancy of c for all observers.
- Simultaneity is relative: events simultaneous in one frame are not simultaneous in another.
- Time dilation: $\Delta t = \gamma \Delta t_0$. Moving clocks run slower. Confirmed by muon survival and atomic clock experiments.
- Length contraction: $L = L_0/\gamma$. Moving objects appear shorter along the direction of motion.
- $E = mc^2$: mass and energy are equivalent. A massive object cannot be accelerated to c ; doing so would require infinite energy.
- Spacetime is a unified four-dimensional continuum; space and time trade off between reference frames while the spacetime interval remains invariant.
- General relativity: gravity is the curvature of spacetime caused by mass and energy. Objects in free fall follow geodesics through curved spacetime.
- Gravitational time dilation: clocks in stronger gravitational fields run slower. GPS systems must correct for this effect.
- Relativity is confirmed by muon decay, GPS corrections, gravitational lensing, gravitational waves, and particle accelerator physics.

Key Terms

special relativity: Einstein's 1905 theory describing physics in inertial (non-accelerating) reference frames; based on the constancy of c and the equivalence of inertial frames.

general relativity: Einstein's 1915 extension of special relativity to include gravity and accelerating frames; describes gravity as curvature of spacetime.

postulate: A foundational assumption accepted as a starting point for a theory; Einstein's two postulates are the basis of special relativity.

reference frame: A coordinate system plus clocks used to assign positions and times to events; an inertial frame is one not undergoing acceleration.

Lorentz factor (γ): $\gamma = 1/\sqrt{1-v^2/c^2}$; the factor by which time intervals are stretched and lengths are contracted due to relative motion; always ≥ 1 .

time dilation: The slowing of a moving clock as measured by a stationary observer; quantified by $\Delta t = \gamma \Delta t_0$.

proper time (Δt_0): The time interval measured by a clock present at both events (i.e., the clock that travels with the moving object); always the minimum time interval.

length contraction: The shortening of a moving object in its direction of motion as measured by a stationary observer; $L = L_0/\gamma$.

proper length (L_0): The length of an object measured in its own rest frame; always the maximum measured length.

simultaneity: Two events occurring at the same time; simultaneity is relative in special relativity — simultaneous events in one frame are not simultaneous in another.

spacetime: The four-dimensional continuum combining three spatial dimensions and one time dimension; the arena of all physical events in relativity.

spacetime interval: A combination of spatial distance and time interval that is the same for all inertial observers; the invariant quantity of special relativity.

equivalence principle: The observation that gravitational and inertial effects are locally indistinguishable; the foundation of general relativity.

geodesic: The straightest possible path through curved spacetime; free-falling objects and light beams follow geodesics.

gravitational time dilation: Clocks in stronger gravitational fields run slower than clocks in weaker fields; a consequence of general relativity.

gravitational lensing: The bending of light paths by the curvature of spacetime near massive objects; confirmed by Eddington in 1919.

gravitational waves: Ripples in spacetime produced by accelerating massive objects; predicted by Einstein in 1916 and first directly detected by LIGO in 2015.

$E = mc^2$: The mass-energy equivalence relation; rest energy equals mass times the square of the speed of light.

Concept Review Questions

Answer in complete sentences.

1. Explain why the Michelson–Morley experiment created a crisis for classical physics. What were physicists expecting to find, and what did they find instead?
 2. State Einstein's two postulates of special relativity in your own words. Which postulate is the most radical departure from classical physics and why?
 3. Why does the constancy of the speed of light force a revision of space and time rather than a revision of electromagnetism?
 4. Explain the relativity of simultaneity using the train and lightning thought experiment. Are both observers right? Explain.
 5. Define proper time. Which observer measures the proper time for an event: the one moving with the clock or the one watching the clock move past?
 6. A spaceship travels at $0.99c$ between two stars. Would the crew age more or less than observers who stay on the departure planet? Explain using time dilation.
 7. Length contraction only affects the dimension along the direction of motion. Why does relativity not also contract perpendicular dimensions?
 8. Explain in conceptual terms why no massive object can ever reach the speed of light. What happens to the energy required as v approaches c ?
 9. What does it mean to say that gravity is the curvature of spacetime rather than a force? Use the rubber sheet analogy and identify its limitations.
 10. Why must GPS satellite clocks be corrected for relativistic effects? Which effect makes them run fast, and which makes them run slow?
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Relativity Thought Experiments

11. You are on a spaceship travelling at $0.6c$ relative to Earth. You shine a flashlight forward. (a) At what speed does the light travel relative to you? (b) At what speed does the light travel relative to Earth? (c) Why don't the speeds add to $1.6c$?
 12. A train 200 m long (proper length) travels at $0.8c$ past a 200 m platform. (a) How long does the train appear to a person on the platform? (b) Does the entire train fit on the platform in the platform's frame? (c) From the train's perspective, how long is the platform? (d) Does the entire train fit on the platform in the train's frame? Explain the apparent contradiction.
 13. Two events occur at the same location and are separated by 5 seconds in one reference frame. (a) Is this a proper time interval or not? Explain. (b) Can any other inertial observer measure a shorter time interval between these events?
 14. Observer A and Observer B move apart at $0.9c$. A emits a light pulse. (a) At what speed does A measure the pulse moving away from her? (b) At what speed does B measure the pulse moving away from A and toward him? (c) What does this tell you about relative simultaneity?
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Time Dilation Problems

15. A muon has a rest-frame half-life of $1.52 \mu\text{s}$. It travels at $0.995c$. (a) Calculate γ . (b) How long is the muon's half-life in the ground frame? (c) How far does the muon travel (in the ground frame) in one half-life?
16. A spaceship travels to a star 40 light-years away at $0.80c$. (a) How long does the trip take in Earth's frame? (b) Calculate γ . (c) How much time elapses on the ship's clocks during the trip? (d) How much has the crew aged?
17. An astronaut travels at $0.995c$ for what she measures as 1.00 year. (a) Calculate γ . (b) How much time passes on Earth? (c) If she returns immediately at the same speed, what is the total Earth time elapsed? (d) How much older is she compared to her twin who stayed on Earth?
18. GPS satellites orbit at 20 200 km altitude and travel at 3.87 km/s . (a) Calculate γ for special relativistic time dilation (use $v = 3870 \text{ m/s}$, $c = 3 \times 10^8 \text{ m/s}$). (b) By how many nanoseconds per day do GPS clocks run slow due to their velocity? (c) GPS clocks actually gain $38 \mu\text{s/day}$ net (after gravitational correction). What does this tell you about the dominant effect?

Spacetime Interpretation Exercises

19. Draw a spacetime diagram (time vertical, position horizontal) showing: (a) a stationary observer at the origin; (b) an observer moving at $0.5c$; (c) a light beam emitted from the origin at $t = 0$. Label each worldline.
20. Event A occurs at position $x = 0$, time $t = 0$. Event B occurs at $x = 3$ light-years, $t = 2$ years. (a) Can A cause B? (b) Is the ordering of A and B the same in all frames? Explain your reasoning using the light travel time between them.
21. Explain in your own words why the spacetime interval is the same for all observers even though individual space and time measurements differ. What does this tell you about the structure of spacetime?
22. A star 100 light-years away explodes today (in Earth's frame). (a) When will we observe the explosion? (b) Did the explosion "just happen" or did it happen long ago? (c) What does this tell you about observing distant events?

Applied Relativity Technology Problems

23. A PET scanner detects two gamma rays from a positron-electron annihilation event. Each photon has energy $E = 511$ keV. (a) Using $E = mc^2$, calculate the mass equivalent of one photon's energy in kg. (b) What was the rest mass of the electron and positron that annihilated? (c) Is all of the rest mass energy converted to photon energy? (electron mass = 9.11×10^{-31} kg)
24. The LHC accelerates protons to $0.999999991c$. (a) Calculate γ . (b) The LHC ring has circumference 27 km. In the proton's rest frame, what is the contracted circumference? (c) How long does it take a proton to complete one lap in the lab frame? (d) How long does it take in the proton's frame?
25. A binary black hole merger event is detected by LIGO. The chirp signal indicates that the orbital frequency just before merger was 150 Hz, and the total mass was 65 solar masses. Using only the concept of gravitational waves as spacetime ripples, explain (qualitatively) what physical quantity LIGO actually measures, and why the sensitivity required (10^{-19} m) is so extraordinary.
26. A spacecraft travelling to Alpha Centauri (4.37 light-years) needs to complete the round trip within the crew's biological lifetime of 50 years (not Earth time). What minimum speed (as a fraction of c) is required? (Set up the time dilation equation and solve for v/c .)

Chapter 51 Checkpoint

Before moving on to Chapter 52, confirm you can do each of the following.

I can...	See Section
Explain why the Michelson–Morley experiment challenged classical physics	51.1
State Einstein's two postulates of special relativity	51.2
Explain why the speed of light is constant in all inertial frames	51.3
Define an inertial reference frame and explain why no frame is preferred	51.4
Explain why simultaneity is relative using a thought experiment	51.5
Calculate time dilation using $\Delta t = \gamma \Delta t_0$ and identify proper time	51.6
Calculate length contraction using $L = L_0/\gamma$ and identify proper length	51.7
Explain $E = mc^2$ and why massive objects cannot reach the speed of light	51.8
Describe spacetime as a unified four-dimensional continuum	51.9
State the equivalence principle and explain how it leads to general relativity	51.10
Describe gravitational lensing and gravitational waves as predictions of GR	51.11
Explain gravitational time dilation and give the GPS example	51.12
Identify common misunderstandings about relativity and correct them	51.14

Continue to Chapter 52: Stars, Galaxies, and Cosmology →